

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JULY / AUGUST 2025

Program	: B.E. – Information Science and Engineering	Semester	: IV
Course Name	: Design and Analysis of Algorithms	Max. Marks	: 100
Course Code	: 21 / 22 / 23IS43	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Describe various asymptotic notations used in analyzing an algorithm. CO1 (09)
Give example for each.
 - With a neat diagram explain the fundamental steps involved in design and analysis of algorithms. CO1 (07)
 - Analyze the following algorithm. CO1 (04)

```

ALGORITHM sum(int n)
{
  if(n==1)
    return n;
  else
    return n* n+ sum(n-1);
}

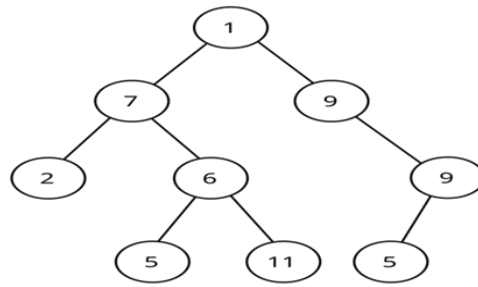
```

 - What does this algorithm compute?
 - What is its basic operation?
 - How many times basic operation is executed?
 - What is the efficiency class of this algorithm?
- Give the general plan for analyzing time efficiency of recursive algorithms. Design a recursive algorithm for Tower of Hanoi and analyze the complexity of the algorithm. CO1 (07)
 - Solve the given Recurrence relations- CO1 (08)
 - $F(n)=F(n-1) + F(n-2)$, $F(0) = 0, F(1) = 1$
 - $T(n)=T(n/2)+c$ where c is constant work per step
 - Derive the time complexity of Linear Search algorithm for best case, average case and worst case efficiency. CO1 (05)

UNIT - II

- Describe Selection Sort and Bubble Sort algorithms. Provide their time complexity. CO2 (08)
 - Write an algorithm for merge sort. Prove that the time complexity of all the three cases (Best, Average and Worst) remains same. CO2 (06)
 - Analyze the time complexity of Binary Search algorithm applied on a given array [12, 22, 45, 67, 74, 81]. Show steps. CO2 (06)
- Write the Quick Sort algorithm. Sort the array [15, 3, 2, 1, 9, 5, 7, 8, 6] in ascending order using Quick Sort and show steps. CO2 (08)

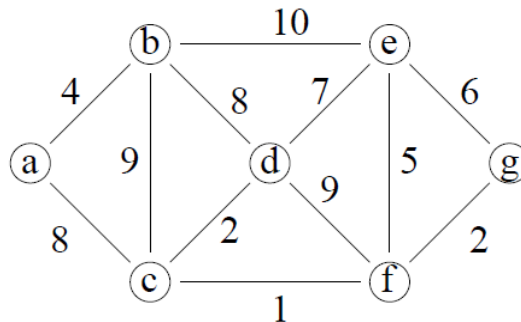
- b) Give the in-order, pre-order, and post-order traversal of the following CO2 (06)
binary tree:



- c) Apply Horspool's string matching algorithm to find the pattern "ARMY" in CO2 (06)
the text "SALUTE TO INDIAN ARMY". Show the shift table and steps.

UNIT - III

5. a) Using Kruskal's algorithm, determine the minimum cost spanning tree CO3 (10)
for the graph given:

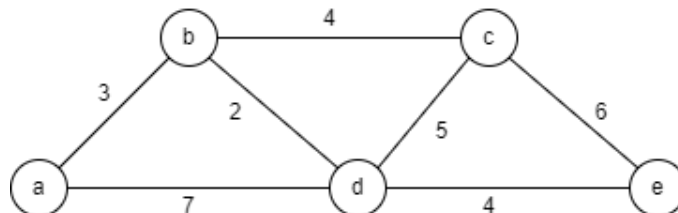


- b) What is Greedy Technique? Construct a Huffman code for the following CO3 (10)
data:

character	A	B	C	D	E	_
probability	0.5	0.35	0.5	0.1	0.4	0.2

- i) Encode the text DAD_BE using the code.
i) Decode the text whose encoding is 1100110110.

6. a) Write Dijkstras algorithm. Apply the same to find shortest paths from CO3 (10)
vertex 'a' to all other vertices of given graph.

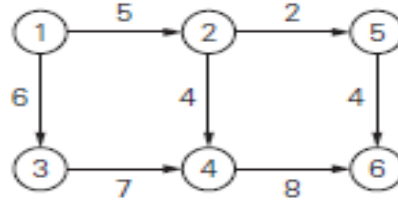


- b) Describe dynamic programing technique. Write a Warshall's algorithm to CO3 (10)
find transitive closure for given adjacency matrix. Apply Warshall's
algorithm to the following adjacency matrix:

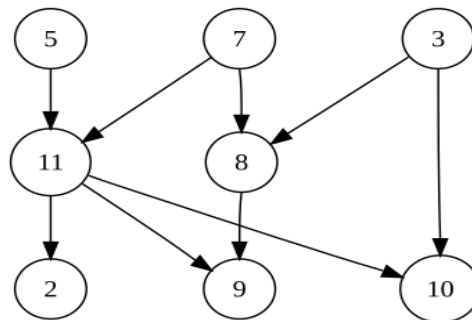
0	1	0	0
0	0	1	0
0	0	0	1
0	1	0	0

UNIT- IV

7. a) Explain Decrease and Conquer strategy using Insertion Sort. Sort [5, 2, 4, 6, 1, 3] step-by-step using insertion sort. CO4 (08)
 b) Perform DFS and BFS on the following graph starting from vertex 1: CO4 (06)



- c) Compare AVL Trees and 2-3 Trees in terms of balancing operations and use cases. CO4 (06)
8. a) Explain AVL Tree rotations with example. Perform insertion of values [10, 20, 30, 40, 50, 25] in an AVL tree and show rotations. CO4 (08)
 b) Construct a 2-3 Tree by inserting the keys: [10, 20, 5, 6, 12, 30, 7, 17]. CO4 (06)
 c) Define Topological sorting. Perform topological sorting on the below given graph: CO4 (06)



UNIT - V

9. a) Build a max-heap from the array: [4, 10, 3, 5, 1]. Perform heapify operations. Then perform Heap Sort on the array. CO5 (08)
 b) Define and differentiate between P, NP, and NP-Complete problems with examples. CO5 (06)
 c) Compare Backtracking and Branch-and-Bound in terms of space efficiency and problem types. CO5 (06)
10. a) Solve the 4-Queens problem using Backtracking. Construct the state space tree for the same. CO5 (08)
 b) Apply Branch-and-Bound method to solve the following 0/1 Knapsack problem: Weights = [2, 3, 4, 5], Values = [3, 4, 5, 6], Capacity = 5. CO5 (08)
 c) Why are NP-Complete problems hard? Discuss consequences for algorithm design. CO5 (04)
